

6th Grade Earth & Space Science — Vocabulary Reference

Georgia Standards of Excellence (S6E1 – S6E6) · Six Unit Vocabulary

Name: _____

Period: _____

Date: _____

Section 1 — The Universe & Solar System

S6E1

Universe: everything that exists: all space, matter, galaxies, and energy.

Galaxy: a huge group of billions of stars held together by gravity.

Milky Way: the galaxy that contains our solar system.

Solar System: the Sun and everything that orbits it: planets, moons, asteroids, and comets.

Star: a giant ball of hot gas that gives off light and heat; the Sun is a star.

Model: a simpler way to represent something that is hard to see or study directly.

Geocentric Model: an old, incorrect model with Earth at the center of the solar system.

Heliocentric Model: the modern model with the Sun at the center of the solar system.

Orbit: the curved path one object takes around another in space.

Gravity: a force that pulls objects toward each other; stronger with more mass.

Inertia: the tendency of a moving object to keep moving in a straight line.

Mass: the amount of matter in an object.

Planet: a large object that orbits the Sun and has cleared its orbit of other objects.

Terrestrial (Rocky) Planet: a small, rocky planet: Mercury, Venus, Earth, or Mars.

Gas Giant: a large planet made mostly of gas: Jupiter, Saturn, Uranus, or Neptune.

Habitable Zone: the 'just right' distance from the Sun where liquid water can exist.

Dwarf Planet: a round object orbiting the Sun that has not cleared its orbit (e.g., Pluto).

Comet: an icy, dusty object that grows a glowing tail when it nears the Sun.

Asteroid: a rocky object that orbits the Sun, mostly in the asteroid belt.

Asteroid Belt: the ring of asteroids between Mars and Jupiter.

Meteoroid: a small rock or piece of metal moving through space.

Meteor: the glowing streak ('shooting star') when a meteoroid burns in the atmosphere.

Meteorite: a piece of space rock that survives and lands on Earth's surface.

Section 2 — The Sun, Earth & Moon

S6E2

Rotation: the spinning of Earth on its axis; one rotation takes about 24 hours (a day).

Revolution: the movement of one object around another; Earth revolves around the Sun in a year.

Axis: the imaginary line through Earth's center that it spins around.

Satellite: an object that orbits a larger object; the Moon is Earth's natural satellite.

Phase: the shape of the lit part of the Moon that we see from Earth.

New Moon: the phase when the lit side faces away from Earth and we see no Moon.

Full Moon: the phase when we see the entire lit side of the Moon.

Waxing: when the lit part of the Moon is getting larger.

Waning: when the lit part of the Moon is getting smaller.

Crescent: a phase when less than half of the Moon appears lit.

Gibbous: a phase when more than half (but not all) of the Moon appears lit.

Eclipse: when the Sun, Earth, and Moon line up so one casts a shadow on another.

Solar Eclipse: when the Moon passes between the Sun and Earth, blocking the Sun.

Lunar Eclipse: when Earth passes between the Sun and Moon, shadowing the Moon.

Tilt: the lean of Earth's axis (about 23.5 degrees); the cause of the seasons.

Season: a part of the year (spring, summer, fall, winter) caused by Earth's tilt.

Hemisphere: one half of Earth, split by the equator into Northern and Southern.

Direct Sunlight: sunlight that hits the ground straight on; concentrated and warm (summer).

Indirect Sunlight: sunlight that hits at a slant; spread out and weaker (winter).

Section 3 — Water on Earth

S6E3

Salt Water: water containing dissolved salt; makes up about 97% of Earth's water.

Fresh Water: water with little or no salt; only about 3% of Earth's water.

Groundwater: fresh water stored underground in soil and rock.

Aquifer: an underground layer of rock or soil that holds groundwater.

Glacier: a large mass of slow-moving ice; stores most of Earth's fresh water.

Water Cycle: the continuous movement of water between the oceans, air, and land.

Evaporation: the change of liquid water into water vapor (gas), powered by the Sun's heat.

Condensation: the change of water vapor back into liquid droplets, forming clouds.

Precipitation: water that falls from clouds as rain, snow, sleet, or hail.

Runoff: water that flows over the land into rivers, lakes, and oceans.

Transpiration: the release of water vapor from plant leaves into the air.

Continental Shelf: the shallow, gently sloping edge of a continent under the ocean.

Continental Slope: the steep drop-off where the shelf ends and the deep ocean begins.

Abysal Plain: the wide, flat floor of the deep ocean.

Mid-Ocean Ridge: a chain of underwater mountains on the ocean floor.

Trench: a deep, narrow valley; the deepest part of the ocean.

Wave: the movement of energy through water; mostly caused by wind.

Current: a steady flow of ocean water in one direction, like a river in the sea.

Tide: the daily rise and fall of the ocean, caused mainly by the Moon's gravity.

Section 4 — Weather & Climate

S6E4

Weather: the condition of the air at one place and time (sunny, rainy, windy).

Climate: the average weather of a place over many years (usually 30+).

Atmosphere: the layer of gases surrounding Earth; mostly nitrogen and oxygen.

Troposphere: the lowest atmospheric layer, where we live and weather happens.

Stratosphere: the atmospheric layer above the troposphere; holds the ozone layer.

Ozone Layer: a part of the stratosphere that absorbs the Sun's harmful UV rays.

Greenhouse Gas: a gas (like carbon dioxide) that traps heat near Earth's surface.

Greenhouse Effect: the trapping of heat by gases in the atmosphere, keeping Earth warm.

Air Pressure: the weight of the air pressing down on a surface.

High Pressure: cool, sinking air; usually brings clear weather.

Low Pressure: warm, rising air; usually brings cloudy, stormy weather.

Wind: moving air, caused by differences in temperature and air pressure.

Sea Breeze: a cool wind that blows from the water toward the land during the day.

Air Mass: a large body of air with similar temperature and moisture throughout.

Front: the boundary where two different air masses meet.

Cold Front: where cold air pushes under warm air; brings sudden, strong storms.

Warm Front: where warm air slides over cold air; brings slow, steady rain.

Thunderstorm: a storm with heavy rain, lightning, and thunder from fast-rising warm air.

Tornado: a violently rotating column of air that reaches the ground.

Hurricane: a large, spinning storm that forms over warm ocean water.

Humidity: the amount of water vapor in the air.

Section 5 — Earth's Surface

S6E5

Crust: Earth's thin, solid, rocky outer layer, where we live.

Mantle: the thick layer of hot, slowly-flowing rock below the crust.

Outer Core: the liquid metal layer (iron and nickel) below the mantle.

Inner Core: the solid metal center of Earth; the hottest layer.

Mineral: a naturally formed solid with a specific chemical makeup (one 'ingredient').

Rock: a solid material made of one or more minerals mixed together.

Igneous Rock: rock that forms when melted rock (magma or lava) cools and hardens.

Sedimentary Rock: rock that forms when layers of sediment are pressed together.

Metamorphic Rock: rock changed by heat and pressure without melting.

Rock Cycle: the slow process by which rocks change from one type into another.

Magma: melted rock beneath Earth's surface.

Lava: melted rock that has reached Earth's surface.

Weathering: the breaking of rock into smaller pieces, in place.

Erosion: the carrying away of weathered rock by water, wind, ice, or gravity.

Deposition: the dropping of eroded material in a new location.

Plate Tectonics: the theory that Earth's crust is broken into moving plates.

Tectonic Plate: a giant piece of Earth's crust that floats on the mantle.

Earthquake: a sudden shaking of the ground when plates slip past each other.

Volcano: an opening in Earth's surface where magma reaches the surface as lava.

Fossil: the preserved remains or trace of a living thing from long ago.

Soil: the loose top layer of weathered rock mixed with organic matter.

Organic Matter: decayed plant and animal material in soil; also called humus.

Bedrock: the solid rock layer beneath the soil.

Section 6 — Natural Resources

S6E6

Natural Resource: anything from the Earth that people use.

Renewable Resource: a resource that refills naturally and won't run out (e.g., sunlight).

Nonrenewable Resource: a resource in limited supply that can run out (e.g., coal).

Sustainable: able to be used without being used up.

Solar Energy: energy from the Sun, captured by solar panels.

Wind Energy: energy from moving air that spins turbines.

Hydroelectric Energy: electricity made from falling or flowing water.

Geothermal Energy: heat energy from inside the Earth.

Biomass: plant material or waste burned to release stored energy.

Tidal Energy: energy from the rise and fall of ocean tides.

Fossil Fuel: a nonrenewable fuel formed from ancient remains: coal, oil, or natural gas.

Coal: a solid fossil fuel formed from buried plant matter.

Oil (Petroleum): a liquid fossil fuel used for gasoline and many products.

Natural Gas: a gaseous fossil fuel used for heating and electricity.

Nuclear Energy: energy released by splitting uranium atoms; uranium is nonrenewable.

Conservation: using resources carefully so they last and stay clean.

Pollution: harmful materials released into the air, water, or soil.

Carbon Dioxide (CO₂): a greenhouse gas released by burning fossil fuels.

Global Temperature Rise: the steady increase in Earth's average temperature over the past century.

Reduce, Reuse, Recycle: the 'Three R's' for using fewer new resources.